

Gender Identification of VR Users by Machine Learning Tracking Data

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ABSTRACT

Gender identification of virtual reality (VR) users by machine learning tracking data could afford personalized experiences, including mitigation of expected human factors or psychology issues. While much research has recently been conducted to identify individual users given their VR tracking data, little research has investigated gender identification. Furthermore, nearly all prior studies have only considered positions and rotations of all the devices. We present a systematic investigation of different combinations and spatial representations of VR tracked devices for predicting a user's gender. Our results indicate head rotations are integral to gender identification while head positions are surprisingly not as important.

Index Terms: Human-centered computing—Virtual Reality; Computing methodologies—Machine learning

1 INTRODUCTION

There are a myriad of potential uses for knowing a virtual reality (VR) user's gender. It has been shown that males and females demonstrate significant differences pertinent to VR, including simulator sickness [11], spatial presence [4], wayfinding [2], color perception [5], empathy [12], and more. Thus, knowing the gender of users would be beneficial in providing personalized VR experiences, such as teleportation-based travel to reduce simulator sickness [1], navigational aids to improve wayfinding [2], and environmental luminance to affect color perception [3].

In recent years, much research has been conducted to explore the feasibility of identifying individual users given their VR tracking data. This has been researched in several contexts, such as watching 360° videos [6], throwing a virtual ball [7], troubleshooting a virtual robot [9], and assembling virtual structures [8]. However, little research has been conducted on identifying the gender of users. Nair et al. [10] recently attempted to identify the gender of VR users playing "Beat Saber" but only achieved 63% accuracy with a transformer model. Tricomi et al. [13] recently investigated identifying the gender of VR users and found F1 scores up to 0.75 at the task level and 0.82 at the action level using logistic regression models.

In this abstract, we present the first systematic investigation of how effectively different combinations of VR tracked devices and their spatial representations can be used to identify the gender of users. For our investigation, we used the public dataset provided by Moore et al. [8] consisting of 45 individuals learning how to assemble two distinct constructions across two sessions. We then systematically investigated seven different combinations of VR tracked devices, seven different spatial representations for those combinations, and three different machine learning models.

Our results indicate that head tracking, particularly head rotation, is important for identifying the gender of users given their VR tracking data, with accuracies up to 91.1%. Surprisingly, we found that head position was not as effective as head rotation. Furthermore, we found that hand tracking data is not effective for identifying

gender. We discuss potential explanations for these results and their implications for VR researchers and developers.

2 MACHINE LEARNING EXPERIMENT

For our systematic investigation and machine learning experiment, we were interested in the following research questions:

- Q1** *How do different combinations of the VR tracked devices affect gender identifiability of users?*
- Q2** *How do different spatial representations of the VR tracked devices affect gender identifiability of users?*

To address these questions, we used the public dataset provided by Moore et al. [8]. This dataset contains the tracking data of 45 participants (20 females, 25 males) performing two different virtual assembly tasks (**task A** and **task B**), for a total of 90 sets of data. Each set provides 16 tracked spatial features (3 position values, 3 Euler angles, 4 quaternion values, and 6D rotation representations) for 3 devices (the headset and two handheld controllers) every frame at 90Hz. These sets of data were segmented into one-second intervals, from which we calculated the maximum, minimum, median, mean, and standard deviation for each of the 48 total spatial features. In our experiment, a gender prediction was made for each one-second interval of data, and then the majority gender classification was chosen as the overall prediction for that participant.

Like Moore et al. [8], we investigated different combinations of the VR tracked devices (**H**: headset, **R**: right hand controller, **L**: left hand controller) at seven levels: H+R+L, H+R, H+L, R+L, H, R, and L. We also investigated their spatial representations (**P**: position, **E**: Euler angles, **Q**: quaternion, and **6D**: 6D representation) at seven levels: P+E, P+Q, P+6D, P, E, Q, and 6D. This yielded a total of 49 conditions to investigate for each transfer task.

We chose to evaluate the same machine learning models as others [6, 8, 9]: gradient boosting machine (**GBM**), k-nearest neighbors (**kNN**), and random forest (**RF**). We used the GBM, kNN, and RF models with default sklearn hyperparameters. Our GBM models had 100 estimators and a learning rate of 0.1. Our kNN models had 5 neighbors ($k = 5$), and the data was normalized to prevent the scale of features from causing an undue effect. Our RF models had 100 estimators. For each model, we used a train/test split: training on all of task A data and then testing on all of task B data, and vice versa.

3 GENDER IDENTIFICATION RESULTS

We reached accuracies of up to 91.1%, as seen in Table 2. We only present the results for our kNN models as they gave us the best overall results but were comparable to our GBM and RF results. **A**→**B** denotes models trained on task A data and tested on task B data while **B**→**A** denotes the opposite.

To answer our research questions, we observe the effects of the tracking devices and their spatial representations on our gender identification accuracies. With regard to tracking devices, the head tracker appears to be the most indicative of gender, as the H+R+L, H+R, H+L, and H conditions yielded the better results. In fact, our best accuracies occur with only the head tracker (H) and no handheld controllers (see Table 2).

In terms of spatial representations, some form of head rotation appears to be critical for accurate gender identification. Surprisingly,

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Table 1: Mean gender identification accuracies from task A to task B with the kNN models across our 49 conditions.

A → B kNN	P+E	P+Q	P+6D	P	E	Q	6D
H+R+L	80.0	75.6	82.2	73.3	86.7	86.7	86.7
H+R	77.8	77.8	80.0	73.3	88.9	84.4	84.4
H+L	80.0	80.0	84.4	68.9	88.9	84.4	84.4
R+L	77.8	80.0	71.1	77.8	66.7	68.9	66.7
H	84.4	82.2	84.4	73.3	82.2	77.8	82.2
R	64.4	68.9	66.7	62.2	68.9	64.4	64.4
L	66.7	68.9	66.7	66.7	55.6	57.8	55.6

Table 2: Mean gender identification accuracies from task B to task A with the kNN models across our 49 conditions.

B → A kNN	P+E	P+Q	P+6D	P	E	Q	6D
H+R+L	75.6	75.6	75.6	68.9	82.2	82.2	84.4
H+R	84.4	82.2	84.4	66.7	84.4	86.7	84.4
H+L	73.3	73.3	75.6	66.7	80.0	77.8	77.8
R+L	68.9	68.9	68.9	62.2	71.1	71.1	68.9
H	86.7	84.4	82.2	80.0	91.1	88.9	91.1
R	60.0	57.8	57.8	57.8	55.6	60.0	60.0
L	57.8	62.2	57.8	57.8	71.1	64.4	62.2

device combinations with only head position (P) and not head rotation performed notably worse than the models with head rotation. Even more surprisingly, our best performing models only considered head rotation (E, Q, or 6D) and not head position.

4 DISCUSSION

Our accuracies reached 91.1%, which is higher than the results achieved in previous attempts at gender identification [10, 13]. This is most likely due to our investigation considering different combinations and spatial representations of the VR devices while most prior works have only considered position and rotation representations of all the devices. This is an especially important distinction as our results indicate that head rotations yield the best gender identification accuracies while head positions only perform worse.

Another important consideration is that we obtained higher accuracies, and for different conditions, than Moore et al. [8], given the same dataset, train/test split, similar conditions, and machine learning models. Their highest reported accuracy was 82.2% for the B→A predictions while ours was 91.1%. Furthermore, their lowest reported accuracy was 15.6% while ours was 55.6%. These results indicate that personal identification is clearly more difficult than gender identification, which is logical given the N classifications compared to the binary classifications. In further comparison to the results of Moore et al. [8], it is interesting to note that head-only (H) models yielded the best results for both personal identification and gender identification. However, even more interesting is the fact that the head rotation only models (HE, HQ, H6D) performed worse than the models including head position for personal identification while we found the opposite results for gender identification. These results

indicate that head rotations are generally the same within gender but different between genders, which may make personal identification of individual users more difficult.

5 CONCLUSION

Gender identification via VR tracking data could afford personalized VR experiences that attempt to mitigate any known issues pertaining to the user's gender. Our results in gender identification reached 91.1% and often had accuracies higher than 80%. These surpass the results found in previous gender identification studies and are higher than prior personal identification accuracies. Further, we found that head rotation is the feature representation most indicative of gender.

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